

INFORMATION AND COMMUNICATION TECHNOLOGY

Assignment 1

NUMBER SYSTEM

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Assignment 1

Number System

Exercise :

If Byte is assumed to be a Number, then what is the range (minimum and maximum value) of Numbers that can be stored in a 8-bit Byte?

The minimum value = "0"

The maximum value = "255"

Exercise :

Represent -10 into 1's complement and 2's complement using 1-Byte.

∴ 1's complement $10 = 1010$

in byte $10 = 00001010$

1's $= 11110101$

So;

$-10 = (11110101)_2$

∴ 2's complement $10 = 1010$

in byte $10 = 00001010$

1's $= 11110101$

2's $= 11110101 + 1$

$= 11110110$

So; $-10 = (11110110)_2$

∴ Reverse Process

1's $= 00001001$

Now add one in it

$= 00001001$

$\begin{array}{r} 1 \\ \hline 00001010 \end{array}$



Exercise

Represent each of the following in 2-bytes using sign magnitude method;

65536

$65536 =$

$$(1000000000000000000)_{\text{2}}$$

Now by adding we need another binary digit.

$$= (01000000000000000000)$$

The total bits we get² is 18

So the solution isn't possible in 2-Bytes.

$$\begin{array}{r} 100 \\ - 32 \\ \hline \end{array}$$

$$-32 =$$

$$(100000000000|00000)$$

2	6 5 5 3 6
2	3 2 7 6 8 $\rightarrow 0$
2	1 6 3 8 4 $\rightarrow 0$
2	8 1 9 2 $\rightarrow 0$
2	4 0 9 6 $\rightarrow 0$
2	2 0 4 8 $\rightarrow 0$
2	1 0 2 4 $\rightarrow 0$
2	5 1 2 $\rightarrow 0$
2	2 5 6 $\rightarrow 0$
2	1 2 8 $\rightarrow 0$
2	6 4 $\rightarrow 0$
2	3 2 $\rightarrow 0$
2	1 6 $\rightarrow 0$
2	8 $\rightarrow 0$
2	4 $\rightarrow 0$
2	2 $\rightarrow 0$
	1 $\rightarrow 0$

2	32
2	16 - 0
2	8 - 0
2	4 - 0
2	2 - 0
	1 - 0

❖ 196

196 =
(0000000011000100)₂

2	196
2	98 - 0
2	49 - 0
2	24 - 1
2	12 - 0
2	6 - 0
2	3 - 0
	1 - 1

❖ -65535

65535 =
(1111111111111111)₂

Now to put significant digit for sign magnitude we need another binary digit (17th Bit) so the solution isn't possible in 2-bytes.

2	65535
2	32767 - 1
2	16383 - 1
2	8191 - 1
2	4095 - 1
2	2047 - 1
2	1023 - 1
2	511 - 1
2	255 - 1
2	127 - 1
2	63 - 1
2	31 - 1
2	15 - 1
2	7 - 1
2	3 - 1
2	1 - 1

* 100015

100015 =
(1100001101010111)₂

Now to put sign
to significant digit
we need to use an
other bit i.e., 18th
Bit so, the solution
isn't possible in
2-Bytes

2	100015	
2	50007	= 1
2	25003	= 1
2	12501	= 1
2	6250	= 1
2	3125	= 0
2	1562	= 1
2	781	= 0
2	390	= 1
2	195	= 0
2	97	= 1
2	48	= 1
2	24	= 0
2	12	= 0
2	6	= 0
2	3	= 0
	1	= 1

Exercise :

Represent the following integer quantities as sequence of bytes encoded using ASCII characters.

* 20456

	2	0	4	5	6
ASCII	50	48	52	53	54

[00110010] [00110000] [00110100] [00110101] [00110110]

÷ 196

<u>1</u>	<u>9</u>	<u>6</u>	
ASCII	49	57	54

00110001, 00111001, 00110110

÷ 1024

<u>1</u>	<u>0</u>	<u>2</u>	<u>4</u>	
ASCII	49	48	50	52

00110001, 00110000, 00110010, 00110100

÷ 32

<u>3</u>	<u>2</u>	
ASCII	51	50

00110011, 00110010

÷ 100015

<u>1</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>5</u>	
ASCII	49	48	48	48	49	53

00110001, 00110000, 00110000, 00110000, 00110001, 00110100